

第二章 函数的导数(Derivative)

K.J.

2.1 导数的定义(The Definition of Derivative)



Multiple Choice Questions

- C 1. Which of the following values is the average rate of change of $f(x) = \sqrt{x+2}$ over the interval $[-1, 2]$?

(A) -4

(B) $-\frac{1}{3}$

(C) $\frac{1}{3}$

(D) 3

- C 2. Let $f(x) = 4 - 3x$. Which of the following is equal to $f'(0)$?

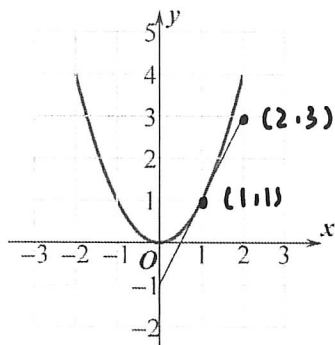
(A) 4

(B) -4

(C) -3

(D) 4

- D 3. The graph of f and its tangent line at $x=1$ is shown below, find $f'(1) =$ (D).



(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) 2

- A 4. Let $f(x) = kx^3 - 2x$, ($k \neq 0$). The average rate of $f(x)$ over the interval $[-2, -1]$ is 3, find k .

(A) $\frac{5}{7}$

(B) $\frac{1}{7}$

(C) $-\frac{1}{7}$

(D) $-\frac{5}{7}$

- C 5. **Calculator** Let $y = \tan e^x - x^2 + 1$. Find $\frac{dy}{dx} \big|_{x=0.5}$

(A) -12.056

(B) 0

(C) 271.066

(D) 273.066

- C 6. **Calculator** The graph of $y = \sqrt{x^2 + 1} - 3x$ crosses the x -axis at one point in the interval $[0, 1]$. What is the slope of the graph at this point?

(A) 0.354

(B) -0.001

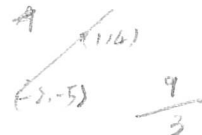
(C) -2.666

(D) 0

- D 7. The population of insects at time t days is modeled by the function P with first derivative $P'(t) = 0.1t^2 + 10t + 120$. Which of the following statements is true?
- (A) At time $t = 10$, the population of insects is 230.
 (B) At time $t = 10$, the population of insects is increasing at a rate of 230 insects per day.
 (C) At time $t = 10$, the population of insects is decreasing at a rate of 230 insects per day.
 (D) At time $t = 10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.

- A 8. If the line tangent to the graph of a function g at the point $(1, 4)$ passes through the point $(-2, -5)$, what is the slope of this line?

- (A) 3 (B) -3 (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$



Free Response Questions

9. Let $f(x) = x^2 - 2x$

(a) Find the average rate of change of f over $[1, 3]$.

(b) Find $\frac{f(2+h) - f(2)}{h}$.

(c) Find the instantaneous rate of change of f at $x = 2$.

$$(a) \frac{f(3) - f(1)}{3 - 1} = \frac{3 - 0}{2} = \frac{3}{2}$$

$$(b) \frac{(2+h)^2 - 2(2+h) - 4 + 4}{h} = \frac{4 + 4h + h^2 - 4 - 2h}{h} = h + 2$$

$$(c) \text{ according to b } f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \text{ at } x = 2. \quad f'(x) = 2.$$

10. Use the definition of derivative to find the derivative of $f(x) = x^3$, then find $f'(2)$.

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^3 - x^3}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{3x^2\Delta x + 3x\Delta x^2 + \Delta x^3}{\Delta x} = \lim_{\Delta x \rightarrow 0} (3x^2 + 3x\Delta x + \Delta x^2) = 3x^2$$

$$f'(2) = 12$$